

Jerry (Yuze) Li

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EDUCATION

University of Waterloo

Fall 2026

BMath, Computer Science | Schulich Leader Scholarship (\$100,000)

Marc Garneau Collegiate Institute (TOPS Program)

Toronto, Canada

SAT: 1570 | AP Scholar with Distinction

Sept. 2022 – June 2026

EXPERIENCE

TopsOJ

July 2023 – Present

Founder & Lead Developer

Toronto, Canada

- Engineered backend evaluation system supporting 120,000+ submissions across 52 contests, serving 2,500+ users across 120+ countries
- Designed EloMMR ranking algorithm for competitive rating; curated and managed a database of 8,500+ problems
- Led 30-member team across development, infrastructure, and sponsorship partnerships including 3Blue1Brown

Quoted Tech Computers

July 2025 – Aug. 2025

Data & Software Engineering Intern

Toronto, Canada

- Developed competitor web scrapers and structured large-scale product datasets for competitive analysis
- Built Node.js + Express backend automating barcode generation and CUPS printing, reducing manual processing time by 70%

Flash Coding

Nov. 2023 – July 2025

Software Engineer Intern

Hybrid

- Built image processing pipelines to extract, crop, and optimize 3,000+ educational resource images
- Developed backend features in PHP and validated REST APIs using Postman

AWARDS

USA Computing Olympiad Platinum Division (Perfect Score on Gold, Jan. 2026)

Journal of High School Science Best Original Paper Award (2026)

AAPT PhysicsBowl Division II – 7th Globally, 1st Nationally

Sir Isaac Newton Physics Contest (SIN) – 9th Globally, 4th Nationally

3x American Invitational Mathematics Examination (AIME) Qualifier

RESEARCH

Computational Astrophysics Research

Nov. 2024 – Present

Student Researcher (advised by Dr. Alexander Mushtukov, University of Oxford)

Remote

- Developed Python simulation framework demonstrating systematic orbital angular momentum loss in binary systems under ballistic mass transfer, contradicting classical $q < 1$ expansion predictions
- First and sole author of “*Ballistic Mass Transfer Necessarily Contracts Orbits in Binary Star Systems*” (Research Notes of the American Astronomical Society, 2026). [10.3847/2515-5172/ae4513](https://doi.org/10.3847/2515-5172/ae4513)
- Independently formulated discrete mass-transfer mappings, deriving closed-form angular momentum drift and introducing a velocity-rescaling closure restoring conservative behavior
- Published sole-authored paper “*On the Conditions Under Which Instantaneous Mass-Transfer Mappings Reproduce Conservative Limits*” (Journal of High School Science, 2026). [10.64336/001c.159153](https://doi.org/10.64336/001c.159153)

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, Java, SQL, HTML, CSS

Frameworks: Flask, Django, Node.js, Express

Libraries: NumPy, Matplotlib, OpenCV

Tools & Systems: Git, Linux, SQLite, Nginx, CUPS, Postman, L^AT_EX